Statistics.

Hypergeometric Distribution

Geometric & Negative Binomial Distributions

Geometric

Again consider a series of Bernoulli's trials (independent trials with constant probability 'p' of a success on each trial. However instead of a fixed no. of trials, trials are conducted until a success is obtained. Let the random variable X denote the no. of trials until the first success.

Definition

In a series of Bernoulli trials (independent trials with constant prob. 'p' of a success), Let the random variable X denote the no. of trials until the first success. The X is a geometric random variable with parameter $0<p<1$ and

$$f(x) = (1-p)^{x-1} p \quad x = 1, 2, \ldots$$

$\mu = E(X) = 1/p$ and $\sigma^2 = V(X) = (1-p)/p^2$.

Negative Binomial Distribution

A generalization of a geometric distribution in which the random variable is the number of Bernoulli trials required to obtain r successes results in the negative binomial distribution.

Lack of Memory Property.

A geometric R.V has been defined as the number of trials until the first success. However, because the trials are independent, the count of the number of trials until the next success can be started at any trial without changing the prob. distn of the random variable. For eg in the transmission of bits, if 100 bits are transmitted, the probability that the first error, after bit 100, occurs on bit 106 is the prob.

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that the next six outcomes are 00000E. This prob is $(0.9)^5(0.1) = 0.059$, which is identical to the prob. that the initial error occurs on bit 6.

The implication of using a geometric model is that the system presumably will not wear out. The prob of an error remains constat for all transmissions. In this sense, the G. Distn is said to lack any memory.

Negative Binomial Distn

In a series of Bernoulli trials (independent trials with constant prob 'p' of a success), let the R.V X denote the no. of trials until 'r' successes occur. Then X is a negative binomial random variable with parameter $0<p<1$ & $r=1,2,3--$ &

$$f(x) = \binom{x-1}{r-1}(1-p)^{x-r}p^r \qquad x = r, r+1, r+2 ----$$

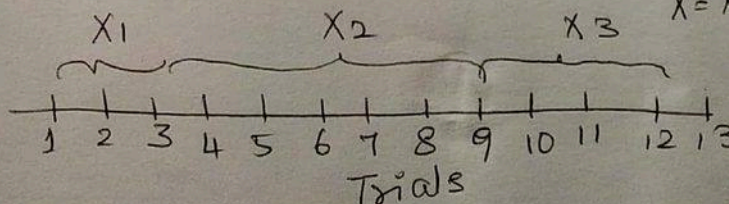
$E(X) = r/p$ & $V(X) = r(1-p)/p^2$.

∵ at least r trials are required to obtain r successes, the of X is from r to ∞, In the sp. case that $r=1$, a -ve binomial r.v is a geometric random variable.

The lack of memory property of a geometric R.V ⇒ the foll.

Let X denote the total no. of trials required to obtain r successes. Let X_1 denot the no of trials reqd to obtain first Succes. Let X_2 denote the no. of extra trials " " " 2nd success
Let X_3 " " " " " " " 3rd success
& so forth. Then the total no. of trials reqd. to obtain r successes is $X = X_1 + X_2 + ---- + X_r$.

A -ve binomial R.V can be interpreted as sum of 'r' geometric r.v.



$X = X_1 + X_2 + X_3$

-ve Binomial R.V rep as sum of geometric R.Vs.

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In -ve binomial the no. of successes is predetermined & the no of trials is random. Hence a -ve binomial r.v can be considered the opp., or negative of binomial r.v.
Binomial R.V is a count of no. of successes in 'n' trials.

Problems.

1. Suppose the R.V 'X' has a geometric distribution with a mean of 2.5. Det. the foll. prob.
   a) $P(X=1)$ b) $P(X=4)$ c) $P(X=5)$ d) $P(X\leq 3)$
   e) $P(X>3)$

$E(X) = 2.5 = 1/p \Rightarrow p = 0.4$.

a) $P(X=1) = (1-0.4)^0 \cdot 0.4 = 0.4$.

b) $P(X=4) = (1-0.4)^3 * 0.4 = 0.0864$.

c) $P(X=5) = (1-0.4)^4 * 0.4 = 0.05184$.

d) $P(X\leq 3) = P(X=1) + P(X=2) + P(X=3)$
$= (1-0.4)^0 * 0.4 + (1-0.4)^1 * 0.4 + (1-0.4)^2 * 0.4$
$= 0.7840$.

e) $P(X>3) = 1 - P(X\leq 3) = 1 - 0.7840 = 0.2160$.

2. The prob. of a successful optical alignment in the assembly of an optical data storage product is 0.8. Assume the trials are independent.
   a) What is the prob. that the first successful alignment requires exactly 4 trials?
   b) What is the prob. that the 1st successful alignment requires at most 4 trials?
   c) What is the prob. that the 1st successful alignment requires at least 4 trials?

Soln Let X denote the no. of trials to obtain 1st successful alignment. Then X is a geometric random variable with $p = 0.8$

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a) $P(X=4) = (1-0.8)^3 * 0.8 = 0.2^3 * 0.8 = 0.0064$.

b) $P(X \leq 4) = P(X=1) + P(X=2) + P(X=3) + P(X=4)$.

$= (1-0.8)^0 0.8 + (1-0.8)^1 0.8 + (1-0.8)^2 * 0.8 + (1-0.8)^3 * 0.8$

$= 0.8 + 0.2 * 0.8 + 0.2^2 * 0.8 + 0.2^3 * 0.8 = 0.9984$.

c) $P(X \geq 4) = 1 - P(X \leq 3) = 1 - P[(X=1)] + P(X=2) + P(X=3)]$

$= 1 - [0.8 + 0.2 * 0.8 + 0.2^2 * 0.8] = 1 - 0.992 = 0.008$.

3. In a clinical study, volunteers are tested for a gene that has been found to increase the risk for a disease. The prob. that a person carries the gene is 0.1.

(a) What is the prob that 4 or more people will have to be tested before 2 with the gene are detected?

(b) How many people are expected to be tested before 2 with the gene are detected?

Let X denote the no of people who carry the gene. Then X is a -ve binomial R.V. with $p=0.1$ and $r=2$.

a) $P(X \geq 4) = 1 - P(X < 4) = 1 - [P(X=2) + P(X=3)]$.

$= 1 - \left[\binom{1}{1}(1-0.1)^0 0.1^2 + \binom{2}{1}(1-0.1)^1 0.1^2\right] =$

$1 - (0.01 + 0.018) = 0.972$.

b) $E(X) = r/p = \frac{2}{0.1} = \underline{\underline{20}}$

4. Assume that each of your calls to a popular radio station has a prob. of 0.02 of connecting, that is, of not obtaining a busy signal. Assume that your calls are independent.

a) What is the prob. that your first call that connects is your tenth call?

b) What is the prob. that it requires > 5 calls for you to connect?

c) What is the mean no. of calls needed to connect?

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Soln Let X denote the no. of calls needed to obtain a connection.

Then X is a geometric R.V with $p = 0.02$.

a. $P(X=10) = (1-0.02)^9 * 0.02 = 0.98^9 * 0.02 = 0.0167$

b. $P(X>5) = 1 - P(X \leq 5) = 1 - [P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)]$

$= 1 - [0.02 + 0.98*0.02 + 0.98^2*0.02 + 0.98^3*0.02 + 0.98^4*0.02]$

$= 1 - 0.0961 = 0.9039$

or can also use no connection in 5 trials (-ve binomial)

$P(X>5) = ({}^5C_0)\,0.02^0 * 0.98^5 = 0.9039$.

c) $E(X) = 1/0.02 = 50$.

5. A fault tolerant system that processes transactions for a financial services firm uses three separate computers. If the operating computer fails, one of the 2 spares can be switched immediately online. After 2nd computer fails, the last computer can be immediately switched online. Assume that the probability of a failure during any transaction is 10^{-8} & the transactions can be considered to be independent events.

a) What is the mean number of transactions before all computers have failed?

b) What is the variance of the no. of transactions before all computers have failed?

Let X denote the no. of transactions until all computers have failed. Then X is -ve binomial R.V with $p = 10^{-8}$ and $r = 3$.

a) $E(X) = 3/10^{-8} = 3*10^8$

b) $V(X) = 3*[1-10^{-8}]/(10^{-8})^2 = 3*10^{16}$.

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6. Hyper Geometric Distribution.

A set of N objects.
K objects classified as successes
N-K " " " failures.
A sample of size n objects is selected randomly (without replacement) from the N objects where $K \leq N$ and $n \leq N$.
Let the random variable X denote the number of successes in the sample. Then X is a hypergeometric random variable and

$$f(x) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}} \qquad x = \max\{0, n+K-N\} \text{ to } \min\{K, n\}$$

The expn. $\min\{K, n\}$ is used in the range of X ∵ the max no. of successes that can occur in the sample is $\leq n$ & the no of successes available, K.

If X is a hypergeometric R.V with parameters N, K & n then.

$$\mu = E(X) = np \quad \& \quad \sigma^2 = V(X) = np(1-p)\left(\frac{N-n}{N-1}\right)$$

where $p = K/N$. (p is interpreted as proportion of successes in the set of N objects)

The term $\frac{N-n}{N-1}$ is called the finite population correction factor.

Problems.

1. A lot of 75 washers contains 5 in which the variability in thickness around the circumference of the washer is unacceptable. A sample of 10 washers is selected at random without replacement.

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a) What is the probability that none of the unacceptable washers is in the sample?

b) What is the prob. that atleast one unacceptable washer is in the sample?

c) What is the prob that exactly one unacceptable washer is in the sample

d) What is the mean no. of unacceptable washers in the sample?

Let X denote the no. of unacceptable washers in the sample of 10.

a) $P(X=0) = \frac{({}^{5}C_{0})({}^{70}C_{10})}{({}^{75}C_{10})} = \frac{\frac{70!}{60!\,10!} * \frac{8!}{0!*8!}}{\frac{75!}{10!\,65!}}$

$= \frac{65*64*63*62*61 * 70*69*68*67*66}{75*74*73*72*71 * 70*69*68*67*66}$

$= 0.4786.$

b) $P(X \geq 1) = 1 - P(X=0) = 1 - 0.4786 = 0.5214.$

c) $P(X=1) = \frac{({}^{5}C_{1})({}^{70}C_{9})}{({}^{75}C_{10})} = \frac{\frac{5!*70!}{9!*61!}}{\frac{75!}{10!*65!}} = 0.3923.$

$= \frac{5*10*65*64*63*62*10}{75*74*73*72*71} = 0.3923.$

d) $E(X) = 10*(5/75) = 2/3.$

2. Printed circuit cards are placed in a final test after being populated with semiconductor chips. A lot contains 140 cards, & 20 cards are selected without replacement for final testing.

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a) If 20 cards are defective, what is the prob. that atleast 1 defective card is in the sample?

b) If 5 cards are defective, what is the prob. that atleast 1 defective card appears in the sample?

Let X denote the no. of cards in the sample that are defective

a) $P(X \geq 1) = 1 - P(X=0)$ $\quad k = 20$ defective. $\quad n = 20$. $\quad N = 140$

$$P(X=0) = \frac{\binom{20}{0}\binom{120}{20}}{\binom{140}{20}}$$

$$= \frac{\frac{20!}{20!*0!} * \frac{120!}{100!*20!}}{\frac{140!}{120!*20!}} = 0.0356.$$

$\therefore P(X \geq 1) = 1 - P(X=0) = 1 - 0.0356 = 0.9644.$

b) $P(X \geq 1) = 1 - P(X=0)$ $\quad K=5.\ n=20\ N=140.$

$$P(X=0) = \frac{\binom{5}{0}\binom{135}{20}}{\binom{140}{20}} = \frac{\frac{135!}{20!*115!}}{\frac{140!}{120!*20!}} = 0.4571.$$

$P(X \geq 1) = 1 - 0.4571 = 0.5429.$

3. Magnetic tape is slit into 1/2" widths that are wound into cartridges. A slitter assembly contains 48 blades. 5 blades are selected at random & evaluated each day for sharpness. If any dull blade is found, the assembly is replaced with a newly sharpened set of blades.

a) If 10 of the blades in the assembly are dull, what is the prob. that the assembly is replaced the 1st day it is evaluated.

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b If 10 of the blades in an assembly are dull, what is the prob that the assembly is not replaced until the 3rd day of evaluation? [Hint: Assume the daily decisions are independent & use the geometric dist.]

c) Suppose on the 1st day of evaluation, 2 of the blades are dull, on the 2nd day " " 6 are dull & on the 3rd day, 10 are dull. What is the prob. that the assembly is not replaced until the 3rd day of evaluation? [Hint: Assume the daily decisions are independent, however the prob. of replacement changes every day]

Let X denote the number of blades in the sample that are dull.

a) $P(X \geq 1) = 1 - P(X=0)$ ← prob not replaced

$$P(X=0) = \frac{\binom{10}{0}\binom{38}{5}}{\binom{48}{5}} = \frac{\frac{38!}{33!*5!}}{\frac{48!}{43!*5!}}$$

K=10 x=0
N-K = 48-10=38
n-x = 5-0

$= 0.2931$

$P(X \geq 1) = 1 - P(X=0) = 0.7069$

b) Let Y denote the no. of days needed to replace the assembly.

$$P(Y=3) = (1-p)^{x-1} * p = (0.7069) * (1-0.7069)^{3-1}$$

Y=3

$= 0.0607$

c) On the 1st day $P(X=0) = \frac{\binom{2}{0}\binom{46}{5}}{\binom{48}{5}} = \frac{\frac{46!}{5!41!}}{\frac{48!}{5!*43!}} = 0.8085$

2nd day $P(X=0) = \frac{\binom{6}{0}\binom{42}{5}}{\binom{48}{5}} = \frac{\frac{42!}{5!37!}}{\frac{48!}{43!*5!}} = 0.4968$

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On the 3rd day for 10 defectives
$P(X=0) = 0.2931$

$\therefore$ P(not replacing until 3rd day) $= 0.8005 * 0.4968 * (1-0.2931)$

$= 0.2811$

Q4. The prob. that your call to a service line is answered in < less than 30 secs is 0.75. Assume that your calls are independent.

a) If you call 10 times, what is the prob. that exactly 9 of your calls are answered within 30 seconds?

b. If u call 20 times, what is the prob. that at least 16 calls are answered in < 30 sec?

c. If u call 20 times, what is the mean no. of calls that are answered in < 30 secs?

d. What is the prob that u must call 4 times to obtain the first answer in < 30 secs?

e. What is the mean no of calls until you are answered < 30 secs?

f. What is the prob. that u must call 6 times in order for ~~you~~ 2 of your calls to be answered in < than 30 secs?

g. What is the mean no. of calls to obtain 2 answers in < 30 secs?

Let X denote the no. of calls that are answered in 30 secs or less.
X is binomial R.V with $p = 0.75$.

a) $P(X=9) = \binom{10}{9}(0.75)^9(0.25)^1 = 0.1877$.

b) $P(X \geq 16) = P(X=16) + P(X=17) + P(X=18) + P(X=19) + P(X=20)$
$= {}^{20}C_{16} * (0.75)^{16}(0.25)^4 + {}^{20}C_{17} * (0.75)^{17} * 0.25^3 +$
${}^{20}C_{18} * (0.75)^{18}(0.25)^2 + {}^{20}C_{19} * (0.75)^{19} * 0.25^1 +$
${}^{20}C_{20} * (0.75)^{20}(0.25)^0 = 0.4148$.

c) $E(X) = 20 * (0.75) = 15$

d) Let Y denote the no. of calls needed to obtain an answer in < 30 secs (Y is geometric R.V)
$P(Y=4) = (1-0.75)^{4-1} * 0.75 = 0.25^3 * 0.75 = 0.0117$.

e. $E(Y) = 1/p = 1/0.75 = 4/3$.

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f. Let W denote the no of calls needed to obtain 2 answers in < 30 secs. Then W has a -ve binomial distn with $p = 0.75$.

$P(W=6) = \binom{5}{1}(0.25)^4(0.75)^2 = 0.0110$.

g. $E(W) = r/p = 2/0.75 = 8/3$.

5. The numbers of messages sent to a computer bulletin board is a Poisson random variable with a mean of 5 msgs per hour.

a) What is the prob. that 5 messages are received in 1hr?

b) " " " " " 10 " " " " 1.5 hrs

c) " " " " " less than 2 msgs are received in 1-half hr?

a) Let X denote the no of msgs sent in 1 hr. $\lambda = 5$

$P(X=5) = \frac{e^{-5}5^5}{5!} = 0.1755$.

b) Let Y denote the no of msgs sent in 1.5 hr. Then Y is poisson R.V with $\lambda = 5*1.5 = 7.5$ msgs/1.5 hr.

$P(Y=10) = \frac{e^{-7.5}(7.5)^{10}}{10!} = 0.0858$.

c) Let W denote the no. of msgs sent in 1-half hour. Then W is a Poisson R.V with $\lambda = 2.5$

$P(W<2) = P(W=0) + P(W=1) = \frac{e^{-2.5}2.5^0}{0!} + \frac{e^{-2.5}2.5^1}{1!}$

$= 0.2878$

6. The no. of errors in a textbook follow a Poisson distn with a mean of 0.01 error/pg. What is the prob that these are 3 or less errors in 100 pgs?

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$X \sim$ (Poisson $\lambda = 0.01$) $\quad X \sim$ Poisson $(\lambda = 1)$ → 102 Pgs

$$P(Y \leq 3) = e^{-1} + \frac{e^{-1}(1)^1}{1!} + \frac{e^{-1}(1)^2}{2!} + \frac{e^{-1}(1)^3}{3!} = 0.9810.$$

1. Det. the pmf for the random variable with the foll cdf

$$F(x) = \begin{cases} 0 & x < 2 \\ 0.2 & 2 \leq x < 5.7 \\ 0.5 & 5.7 \leq x < 6.5 \\ 0.8 & 6.5 \leq x < 8.5 \\ 1 & 8.5 \leq x \end{cases}$$

$$\text{pmf} = P_X(x) = \begin{cases} 0.2 & x = 2 \\ 0.3 & x = 5.7 \\ 0.3 & x = 6.5 \\ 0.2 & x = 8.5 \\ 0 & \text{otherwise} \end{cases}$$

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PROBABILITY DISTRIBUTIONS AND PROBABILITY DENSITY FUNCTIONS

A probability density function $f(x)$ can be used to describe the probability distribution of a continuous random variable X. If an interval is likely to contain a value for X, its probability is large and it corresponds to large values for $f(x)$. The prob. that X is between a and b is determined as the integral of $f(x)$ from a to b.

For a continuous random variable X, a probability density function is a function such that

1) $f(x) \geq 0$.
2) $\int_{-\infty}^{\infty} f(x)\,dx = 1$.
3) $P(a \leq X \leq b) = \int_{a}^{b} f(x)\,dx$ = area under $f(x)$ from a to b. for any a and b.

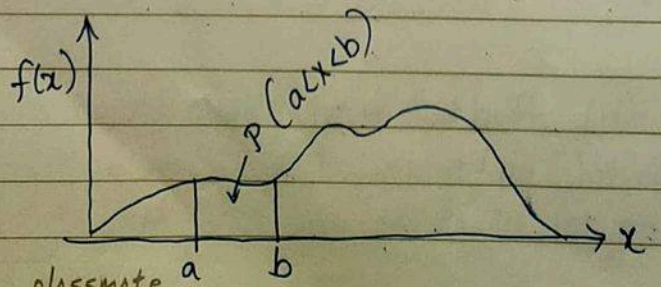
A pdf is zero for x values that cannot occur and it is assumed to be zero wherever it is not specifically defined.

A histogram is an approx. to pdf

For a continuous random variable X and any value x
$P(X = x) = 0$. The pdf at any pt is zero.

If X is a continuous r.v, for any x_1 and x_2

$$P(x_1 \leq X \leq x_2) = P(x_1 < X \leq x_2) = P(x_1 \leq X < x_2) = P(x_1 < X < x_2) \quad \text{---------- (1)}.$$



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Problems:

1. Suppose that $f(x) = e^{-(x-4)}$ for $4 < x$. Det the foll. prob.

a) $P(1<X)$ b) $P(2 \le X \le 5)$ c) $P(5<X)$ d) $P(8<X<12)$

e) Det x such that $P(X<x) = 0.90$.

a) $P(1<X) = \int_4^{\infty} e^{-(x-4)}\,dx = -e^{-(x-4)}\Big|_4^{\infty} = 1 \quad \because f_x(x) = 0 \text{ for } x < 4$

This can also be obtained from the fact that $f_x(x)$ is pdf for $4<x$.

b) $P(2 \le X < 5) = \int_4^5 e^{-(x-4)}\,dx = -e^{-(x-4)}\Big|_4^5 = 1 - e^{-1}$

$= 0.6321$

c) $P(5<X) = 1 - P(X \le 5)$, from b $P(X \le 5) = 0.6321$

$\therefore P(5<X) = 0.3679$.

d) $P(8<X<12) = \int_8^{12} e^{-(x-4)}\,dx = -e^{-(x-4)}\Big|_8^{12} = e^{-4} - e^{-8} = 0.0180$.

e. $P(X<x) = \int_4^x e^{-(x-4)}\,dx = -e^{-(x-4)}\Big|_4^x = 1 - e^{-(x-4)} = 0.90$.

$\Rightarrow x = 4 - \ln(0.10) = 6.303$

2) Suppose that $f(x) = x/8$ for $3 < x < 5$. Det. the foll prob.

a) $P(X<4)$ b) $P(X>3.5)$ c) $P(4<X<5)$ d) $P(X<4.5)$

e) $P(X<3.5 \text{ or } X>4.5)$.

a) $P(X<4) = \int_3^4 x/8\,dx = \frac{x^2}{2*8}\Big|_3^4 = \frac{4^2}{2*8} - \frac{3^2}{8*2}$

$= 1 - \frac{9}{16} = \frac{7}{16} = 0.4375$.

b. $P(x>3.5) = \int_{3.5}^5 \frac{x}{8}\,dx = \frac{x^2}{16}\Big|_{3.5}^5$

$= \frac{5^2}{16} - \frac{(3.5)^2}{16} = 0.7969$.

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c. $P(4<x<5) = \int_4^5 \frac{x}{8} = \frac{x^2}{16}\Big|_4^5 = \frac{25}{16} - \frac{16}{16} = \frac{9}{16}$

$= 0.5625$.

d) $P(x<4.5) = \int_3^{4.5} \frac{x}{8} = \frac{x^2}{16}\Big|_3^{4.5} = \frac{11.25}{16} = 0.7031$

e) $P(x<3.5 \text{ or } x>4.5)$

$= \int_3^{3.5} \frac{x}{8}dx = \frac{x^2}{16}\Big|_3^{3.5} = 0.20315$.

$+ \int_{4.5}^5 \frac{x}{8}dx = \frac{4.75}{16} = 0.296875$

$= 0.500$

3. Suppose that $f(x) = e^{-x}$ for $0<x$. Det foll prob

a) $P(1<x)$ b) $P(1<X<2.5)$

c) $P(X=3)$ d) $P(X<4)$ e) $P(3\leq x)$

f) Det x $P(x<X) = 0.10$

g) " " $P(X\leq x) = 0.10$.

$f(x) = e^{-x}$ for $0<x$.

a. $P(1<x) = \int_1^\infty e^{-x}dx = -e^{-x}\Big]_1^\infty = -e^{-\infty} + e^{-1} = 0.3679$

b. $P(1<x<2.5) = \int_1^{2.5} e^{-x}dx = -e^{-x}\Big]_1^{2.5} = -e^{-2.5} + e^{-1} = 0.2858$

c. $P(x=3) = 0$.

d. $P(x<4) = \int_0^4 e^{-x}dx = -e^{-x}\Big]_0^4 = -1 + 0.018315$

$= 0.9816$.

e. $P(3\leq x) = \int_3^\infty e^{-x}dx = -e^{-3} + e^{-\infty}$

$= 0.0498$.

f. $P(x<X) = 0.1 = \int_x^\infty e^{-x}dx = -e^{-x}\Big]_x^\infty$

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$= -e^{-x} = 0.1$ $\boxed{x = 2.30}$

q. $P(X \leq x) = 0.10 = \int_0^x e^{-x} dx = -e^{-x}\Big]_0^x = 0.1$

$= -e^{-0} + e^{-x} = 0.1$

$\Rightarrow -1 + e^{-x} = 0.1 \Rightarrow e^{-x} = 0.9 \Rightarrow \ln 0.9 = -x$

$\boxed{x = 0.105}$

CDF- The cumulative distribution fn. of a continuous random variable X is

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(u)\, du.$$

1. Suppose the cdf of random variable X is

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.2x & 0 \leq x < 5 \\ 1 & 5 \geq x \end{cases}$$

Det. the foll. a) $P(X < 2.8)$ b) $P(X > 1.5)$
c) $P(X < -2)$ d) $P(X > 6)$.

1. $P(X < 2.8) = 0.2 * x = 0.2 * 2.8 = 0.56$.

2. $P(X > 1.5) = 1 - P(X < 1.5) = 1 - 0.2 * 1.5 = 1 - 0.3 = 0.7$.

3. $P(X < -2) = 0$

4. $P(X > 6) = 0$.

2. CDF of a random variable X is

$$F(x) = \begin{cases} 0 & x < -2. \\ 0.25x + 0.5 & -2 \leq x < 2. \\ 1 & 2 \leq x \end{cases}$$

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Det. the foll.

a $P(X<1.8)$ b) $P(X>-1.5)$ c) $P(X<-2)$

d $P(-1<X<1)$

a $P(X<1.8) = 0.25*1.8 + 0.5 = 0.95$.

b. $P(X>-1.5) = 1 - P(X<-1.5) = 1 - (0.25*-1.5 + 0.5) = 0.875$

c. $P(X<-2) = 0$

d. $P(-1<X<1) = P(X<1) - P(X>-1)$

$= 0.25*1 + 0.5 - (0.25*-1 + 0.5)$

$= 0.75 - 0.25 = 0.5$

3. The gap width is an imp property of a magnetic recording head. In coded units, if the width is a continuous random variable over a range from $0<x<2$ ~~width~~ with $f(x) = 0.5x$, det the cum. distribution fn. of the gap width

$$F(x) = \begin{cases} 0 & x<0 \\ 0.25x^2 & 0<x<2 \\ 1 & 2 \leq x \end{cases}$$

Mean and Variance of a continuous random variables

For a continuous random variable with pdf $f(x)$ the mean or expected value of X, denoted as μ or $E(X)$ is

$$\mu = E(X) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx.$$

Variance of X denoted as $V(X)$ or σ^2 is

$$\sigma^2 = V(X) = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx = \int_{-\infty}^{\infty} x^2 f(x)\, dx - \mu^2$$

Std dev of X is $\sigma = \sqrt{\sigma^2}$.

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1. Suppose $f(x) = 0.25$ for $0 < x < 4$. Det the mean & variance of X.

$$E(X) = \int_0^4 x \cdot 0.25\, dx = 0.25 \frac{x^2}{2}\Big|_0^4$$

$$= \frac{0.25 \times 16}{2} - 0.25 \times 0 = \underline{\underline{2}}$$

$$V(X) = \int_0^4 x^2 * 0.25\, dx - 2^2$$

$$= \frac{0.25 x^3}{3}\Big|_0^4 - 4 = \frac{64}{3} * 0.25 - 4$$

$$= \frac{16-12}{3} = \frac{4}{3}$$

2. The thickness of a conductive coating in micrometers has a density fn. of $600x^{-2}$ for $100\,\mu m < x < 120\,\mu m$.

a) Det the mean and variance of the coating thickness

b) If the coating costs $\$0.50/\mu m$ of thickness on each part, what is the average cost of coating per part.

$$E(X) = \int_{100}^{120} x \cdot \frac{600}{x^2}\, dx = \int_{100}^{120} \frac{600}{x}\, dx$$

$\int x^n dx = \frac{x^{n+1}}{n+1} + C$ here $n = -1$ $\therefore \int x^{-1} dx = \frac{x^{-1+1}}{-1+1} = \ln(x) + C$

$$\therefore \int_{100}^{120} \frac{600}{x} = 600 * (\ln x)\Big|_{100}^{120}$$

$$= 600 * [\ln(120) - \ln(100)] = 109.39$$

$$V(X) = \int_{100}^{120} x^2 * \frac{600}{x^2}\, dx - \mu^2$$

$$= \int_{100}^{120} 600\, dx - (109.39)^2 = 600x\Big|_{100}^{120} - (109.39)^2$$

$$= 12000 - 11966.1721 = 33.8279.$$

Total cost = mean * cost / μm = 109.39 * 0.5

= 54.70$.

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3. Suppose the pdf of the length of computer cables is $f(x) = 0.1$ from 1200 to 1210 mm.

a) Det. the mean and std. devn. of the cable length.

b) If the length specifications are $1195 < x < 1205$ mm, what proportion of cables are within specifications.

a) $$E(X) = \int_{1200}^{1210} x * 0.1\,dx = 0.05x^2 \Big|_{1200}^{1210} = 1205.$$

$$V(X) = \int_{1200}^{1210} (x-1205)^2 * 0.1\,dx = 0.1 \frac{(x-1205)^3}{3} \Big|_{1200}^{1210} = 8.333$$

$\therefore V(X) = 8.333 \Rightarrow \sigma = \sqrt{8.333} = 2.887.$

b) Clearly, centering the process at the center of the specfns. results in the greatest proportion of cables within specfns.

$$P(1195 < X < 1205) = P(1200 < X < 1205) = \int_{1200}^{1205} 0.1\,dx = 0.1x \Big|_{1200}^{1205}$$

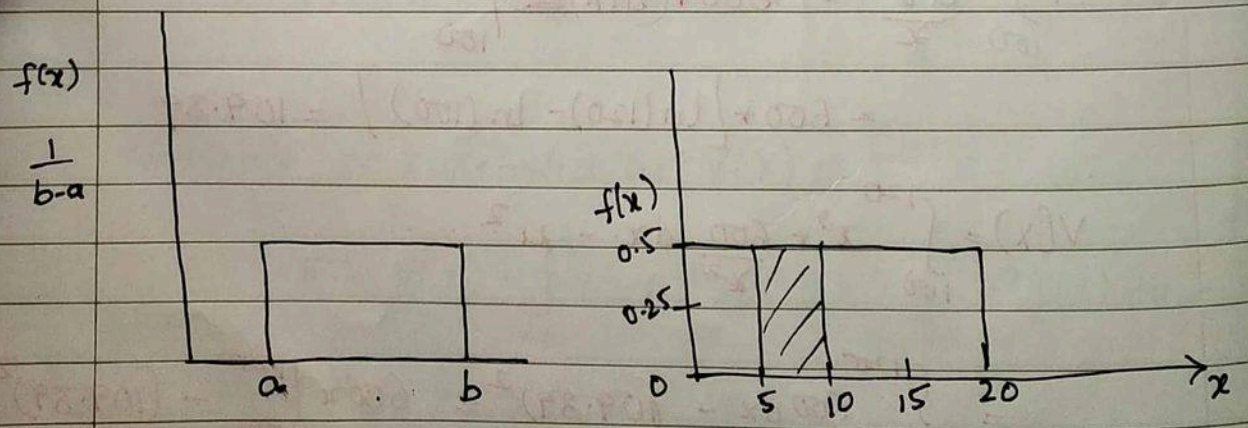
$$= 0.5$$

CONTINUOUS UNIFORM DISTRIBUTION.

Defn. A continuous random variable X with pdf

$$f(x) = 1/(b-a) \quad a \le x \le b$$

is a continuous uniform random variable.



Continuous UNIFORM PDF

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The mean of continuous uniform R.V is

$$E(X) = \int_a^b \frac{x}{b-a}\,dx = \frac{0.5x^2}{b-a}\Big|_a^b = \left(\frac{a+b}{2}\right)$$

Variance of X is

$$V(X) = \int_a^b \frac{\left(x-\left(\frac{a+b}{2}\right)\right)^2}{b-a}\,dx = \frac{\left(x-\frac{a+b}{2}\right)^3}{3(b-a)}\Big|_a^b = \frac{(b-a)^2}{12}$$

$$\boxed{\mu = \frac{a+b}{2} \qquad V(X) = \sigma^2 = \frac{(b-a)^2}{12}}$$

$$F(x) = \int_a^b 1/(b-a)\,dx = \frac{x}{(b-a)} - \frac{a}{(b-a)} = \left(\frac{x-a}{b-a}\right) \qquad F(x) = \begin{cases} 0 & x<a \\ (x-a)/b-a & a\le x<b \\ 1 & b\le x \end{cases}$$

4-33 1. The net weight in pounds of a packaged chemical herbicide is uniform for $49.75 < x < 50.25$ pounds.

a) Det the mean & variance of the wt of packages.

b) Det the cdf of weight of the packages.

c) Det. $P(X<50.1)$

Let X be a random variable with continuous uniform distn. over the interval $[49.75, 50.25]$ pounds with pdf.

$$f(x) = 1/b-a = 1/0.5 = 2, \quad 49.75 < x < 50.25.$$

a) Mean $E(X) = \frac{a+b}{2} = \frac{49.75+50.25}{2} = \underline{\underline{50}}$

$$Var(X) = \frac{(50.25-49.75)^2}{12} = 0.0208 \qquad \sigma^2 = \sqrt{0.0208} = 0.144.$$

b. $$f(x) = \begin{cases} 0 & x \le 49.75. \\ \frac{x-49.75}{50.25-49.75} = 2x-99.5 & 49.75 \le x < 50.25. \\ 1 & x \ge 50.25. \end{cases}$$

c $P(X<50.1) = F(50.1) = 2*50.1 - 99.5 = \underline{\underline{0.7}}$

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4.34. The thickness of a flange on an aircraft component is uniformly distributed b/w 0.95 & 1.05 mm.

a) Det. cdf of the flange thickness.

b) Det. the proportion of flanges that exceeds 1.02 mm.

c) What thickness is exceeded by 90% of the flanges?

d) Det. the mean & variance of flange thickness.

The pdf of fn $a = 0.95$ $b = 1.05$

$f(x) = 1/0.1 = 10$ for $0.95 < x < 1.05$.

a) $$F(x) = \begin{cases} 0 & x < 0.95 \\ \frac{x - 0.95}{1.05 - 0.95} = 10x - 9.5 & 0.95 \le x < 1.05 \\ 1 & 1.05 \le x \end{cases}$$

b) $P(X > 1.02) = 1 - P(X \le 1.02) = 1 - F_X(1.02) = 1 - (10 * 1.02 - 9.5)$

$= 1 - 0.7 = 0.3$

c) If $P(X > x) = 0.9$ then $1 - F(x) = 0.90$ and $F(x) = 0.10$.

$\therefore 10x - 9.5 = 0.10$ & $x = 0.96$.

d) $E(X) = (1.05 + 0.95)/2 = 1.00$.

$V(X) = \frac{(1.05 - 0.95)^2}{12} = 0.00083$.

4.35 Suppose the time it takes a data colln. operator to fill out an electronic form for a database is uniformly distributed b/w 1.5 & 2.2 mins.

a) What is the mean & variance of the time it takes an operator to fill out the form?

b) What is the prob. that it will take < 2 mins to fill out the form?

c) Det the cdf of the time it takes to fill out the form.

$E(X) = (1.5 + 2.2)/2 = 1.85$ min.

$V(X) = \frac{(2.2 - 1.5)^2}{12} = 0.0408$ mins^2

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b. $P(X<2) = \int_{1.5}^{2} \frac{1}{(2.2-1.5)}\,dx = \int_{1.5}^{2} \frac{1}{0.7}\,dx = \frac{1}{0.7}x \Big|_{1.5}^{2}$

$= \frac{2-1.5}{0.7} = \frac{0.5}{0.7} = 0.7143$

c) $F(x) = \int_{1.5}^{x} \frac{1}{(2.2-1.5)}\,dx$

$$F(x) = \begin{cases} 0 & x < 1.5 \\ \frac{x-1.5}{0.7} & 1.5 \le x < 2.2 \\ 1 & 2.2 \le x \end{cases}$$

4.37. The thickness of photoresist applied to wafers in semiconductor manufacturing at a particular location on the wafer is uniformly distributed b/w 0.2050 & 0.2150 micrometers.

a) Det the cdf of photoresist thickness.

b) Det the proportion of wafers that exceeds 0.2125 μm in photoresist thickness.

c) What thickness is exceeded by 10% of the wafers?

d) Det. the mean & variance of photoresist thickness.

pdf is $f(x) = \frac{1}{b-a} = \frac{1}{0.2150-0.2050} = \frac{1}{0.01} = 100$. for $0.2050 < x < 0.2150$.

a. $\therefore$ cdf $= \begin{cases} 0 & x < 0.2050 \\ \frac{x - 0.2050}{0.0100} = 100x - 20.50 & 0.2050 \le x < 0.250 \\ 1 & 0.2150 \le x \end{cases}$

b. $P(X > 0.2125) = 1 - F(0.2125) = 1 - [100(0.2125) - 20.50] = 0.25$.

c) If $P(X > x) = 0.10$ then $1 - F(x) = 0.10 \Rightarrow F(x) = 0.90$.

$\therefore 100x + 20.50 = 0.90 \Rightarrow x = 0.2140$.

d) $E(X) = 0.2100\ \mu m$.

$V(X) = 8.33 * 10^{-6}\ \mu m^2$.

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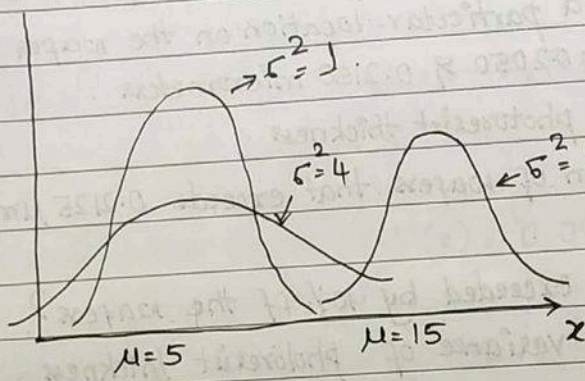
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NORMAL DISTRIBUTION.

Most widely used model for distn of a random variable is a normal distribution.

Whenever a random expt. is replicated, the r.v that equals the avg (or total) result over the replicates tends to have a normal distn. as the no. of replicates become large. De Moivre presented this fundamental result, known as the central limit theorem in 1733. After 100yrs Gauss independently developed a normal distn. A normal distn is also referred to as a Gaussian distribution.



R.V with diff means and variances can be modeled by normal pdf with appropriate choices of the center & width of the curve. The value of $E(X) = \mu$ determine the center of the pdf & the value of $V(X) = \sigma^2$ determines the width.

A random variable X with pdf

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}} \quad -\infty < x < \infty.$$

is a normal random variable with parameters μ where $-\infty < \mu < \infty$ & $\sigma > 0$.

Also $E(X) = \mu$ & $V(X) = \sigma^2$.

& the notation $N(\mu, \sigma^2)$ is used to denote the distn

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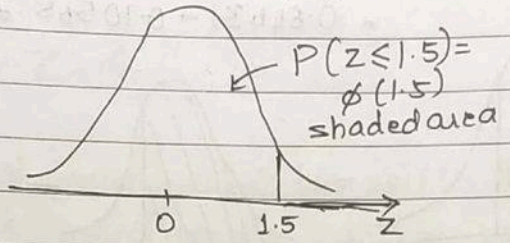
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A normal random variable with
$\mu = 0$ and $\sigma^2 = 1$.
is called a standard normal random variable & is denoted as Z.

The cum. distn. fn. of a std. normal r.v is denoted as
$\phi(z) = P(Z \leq z)$.

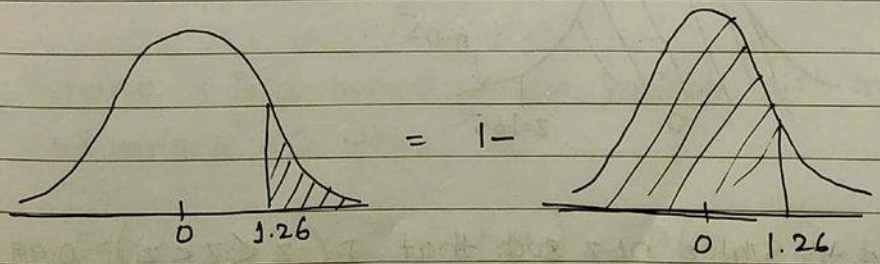
Std. normal prob. density function



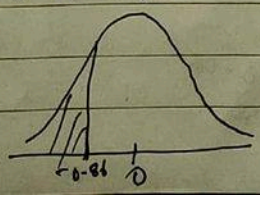
z	0.00	0.01	0.02	0.03
0	0.5	0.50399	0.50398	0.51197
⋮		⋮		
1.5	0.93319	0.93448	0.93574	0.93699

Prob that are not of the form $P(Z \leq z)$ are found by using basic rules of prob & the symmetry of normal distn. along with Tables.

Ex. 1) $P(Z > 1.26) = 1 - P(Z \leq 1.26) = 1 - 0.89616 = 0.10384$



2) $P(Z < -0.86) = 0.19490$



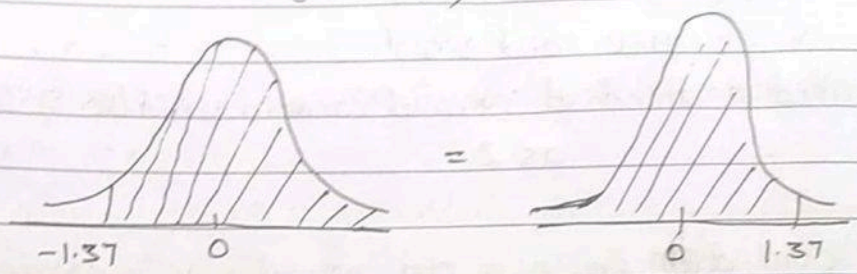
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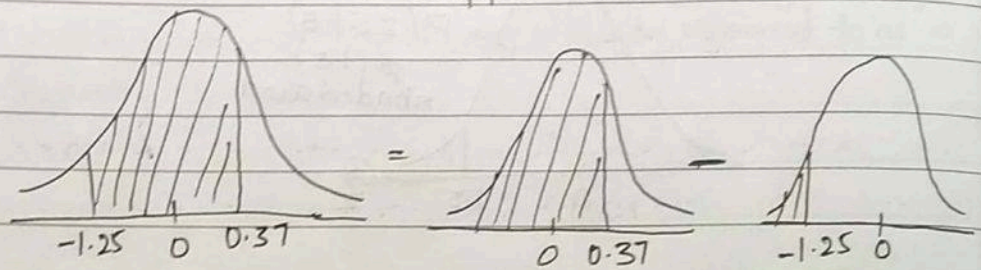
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3. $P(z > -1.37) = P(z < 1.37) = 0.91465$



4. $P(-1.25 < z < 0.37) = P(z < 0.37) - P(z < -1.25)$

$= 0.64431 - 0.10565 = 0.53866.$

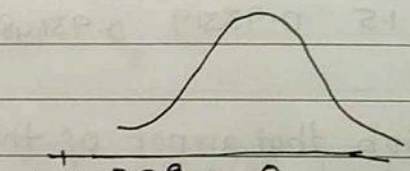


5. $P(z \leq -4.6)$ cannot be found exactly from Table.

The last value in table is $P(z \leq -3.99) = 0.00003$

$P(z \leq -4.6) < P(z \leq -3.99)$

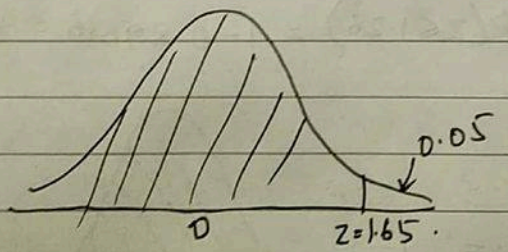
$P(z \leq -4.6) = 0.$



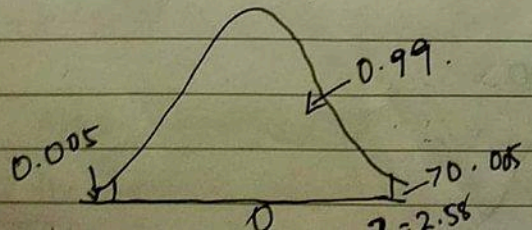
6) $P(Z > z) = 0.05$

$= P(z \leq z) = 0.95.$

nearest value $= 0.95053$, $z = 1.65.$



7) Find the value of z such that $P(-z < z < z) = 0.99.$



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$P(-z < Z < z) = 0.99$.

∵ of the symmetry of normal distn, for the area 0.99, the area in each tail of the distn must equal 0.005.

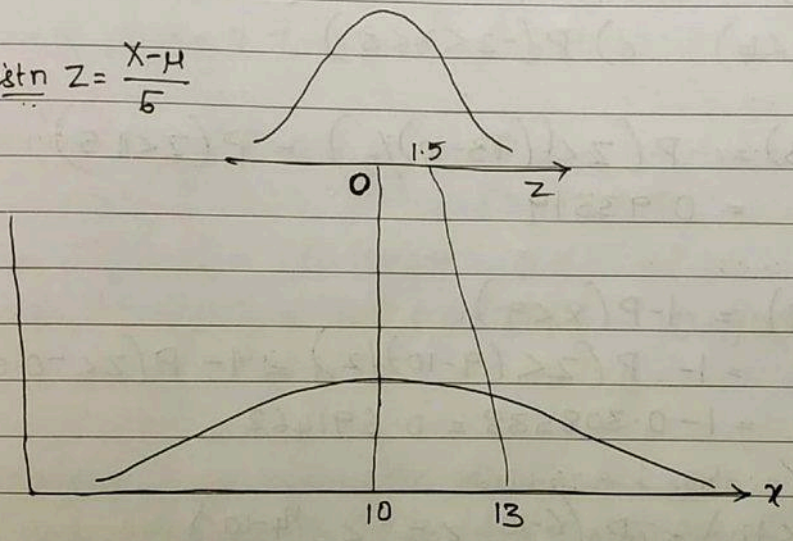
∴ the value for z corresponds to a prob. of 0.995 in table. The nearest prob. from tables is 0.99506 when $z = 2.58$

If X is a normal random variable with $E(X) = \mu$ and $V(X) = \sigma^2$, the random variable

$$Z = \frac{X - \mu}{\sigma}$$

is a normal random variable with $E(Z) = 0$ and $V(Z) = 1$, That is Z, is a std normal random variable.

Distn $Z = \frac{X-\mu}{\sigma}$



Standardizing a normal random variable.

Suppose X is a normal random variable with mean μ and variance σ^2. Then,

$$P(X \leq x) = P\left(\frac{X-\mu}{\sigma} \leq \frac{x-\mu}{\sigma}\right) = P(Z \leq z)$$

where Z is a std normal r.v & $z = \frac{(x-\mu)}{\sigma}$ is the z-value obtained by standardizing X.

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Problems.

4.41 1. Use tables to det. the foll prob. for std. normal r.v. z,

a) $P(z<1.32)$ b) $P(z<3.0)$ c) $P(z>1.45)$

d) $P(z>-2.15)$ e) $P(-2.34<z<1.76)$

a. $P(z<1.32) = 0.906582$.

b. $P(z<3.0) = 0.99865$.

c. $P(z>1.45) = 1 - P(z<1.45) = 1 - 0.92647 = 0.07353$.

d) $P(z>-2.15) = P(z<2.15) = 0.98422$.

e) $P(-2.34<z<1.76) = P(z<1.76) - P(z>2.34)$

$= 0.9607 - (1 - P(z<2.34))$

$= 0.9607 - 0.009642 = 0.9511$ (0.990358)

2. Assume X is normally distributed with a mean of 10 & a std devn of 2. Det. the foll.

a) $P(X<13)$ b) $P(X>9)$ c) $P(6<X<14)$

d) $P(2<X<4)$ e) $P(-2<X<8)$

a. $P(X<13) = P(z<((13-10)/2) = P(z<1.5)$

$= 0.93319$.

b. $P(X>9) = 1 - P(X<9)$

$= 1 - P(z<(9-10)/2) = 1 - P(z<-0.5)$

$= 1 - 0.308538 = 0.691462$.

c) $P(6<X<14) = P\left(\frac{6-10}{2} < z < \frac{14-10}{2}\right)$

$= P(-2<z<2)$

$= [P(z<2) - P(z<-2)]$

$= 0.977250 - 0.022750 = 0.954500$.

d) $P(2<X<4) = P\left(\frac{2-10}{2} < z < \frac{4-10}{2}\right)$

$= P(-4<z<-3)$

$= P(z<-3) - P(z<-4)$

$= 0.001350 - 0.000033 = 0.001317$

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e. $P(-2<X<8) = P(X<8) - P(X<-2)$

$= P\left(Z < \frac{8-10}{2}\right) - P\left(Z < \frac{-2-10}{2}\right)$

$= P(Z<-1) - P(Z<-6) = 0.15866$

4.43. Assume Z has a std normal distn. Use table to det. the value for z that solves each of the foll

a. $P(Z<z) = 0.9$.

$P(Z<1.28) = 0.9$.

b. $P(Z<z) = 0.5$. $P(Z<0) = 0.5$.

c) $P(Z>z) = 0.1 \Rightarrow P(Z<z) = 0.9$ & $z = 1.28$.

d $P(Z>z) = 0.9 \Rightarrow P(Z<z) = 0.10$ & $z = -1.28$.

e) $P(-1.24 < Z < z) = 0.8$

$\Rightarrow P(Z<z) - P(Z<-1.24) = 0.8$.

$= P(Z<z) - 0.10749 = 0.8$

$\Rightarrow P(Z<z) = 0.8 + 0.10749 = 0.90749$.

$z = 1.33$.

4.46. Assume X is normally distributed with a mean of 10 & a std. devn. of 2. Det. the value for x that solves each of the foll.

a) $P(X>x) = 0.5 \Rightarrow P\left(Z > \frac{x-10}{2}\right) = 0.5$.

$\therefore \frac{x-10}{2} = 0$ and $x = 10$.

b. $P(X>x) = 0.95 = P\left(Z > \frac{x-10}{2}\right) = 1 - P\left(Z < \frac{x-10}{2}\right)$

$= 0.95$.

$\therefore P\left(Z < \frac{x-10}{2}\right) = 0.05$ & $\frac{x-10}{2} = -1.64$

$x = 6.72$.

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c) $P(x<X<10) = 0.2$.

$P\left(\frac{x-10}{2} < Z < 0\right) = P(Z<0) - P\left(Z < \frac{x-10}{2}\right)$

$= 0.5 - P\left(Z < \frac{x-10}{2}\right) = 0.2$.

$\therefore P\left(Z < \frac{x-10}{2}\right) = 0.3 \Rightarrow \frac{x-10}{2} = -0.52$.

$x = 8.96$.

4.47/4.49. The compr. strength of samples of cement can be modeled by a normal distn. with a mean of 6000 kgs/cm^2 & std. devn. of 100 kgs/cm^2.

a) What is the prob that a samples strength is less than 6250 kg/cm^2?

b) What is the prob that a sample strength is b/w 5800 & 5900 kg/cm^2.

c. What strength is exceeded by 95% of the samples?

a. $P(X < 6250) = P\left(Z < \frac{6250-6000}{100}\right)$

$= P(Z < 2.5) = 0.99379$.

b $P(5800 < X < 5900) =$

$P\left(\frac{5800-6000}{100} < Z < \frac{5900-6000}{100}\right)$

$= P(-2 < Z < -1) = P(Z<-1) - P(Z<-2)$

$= 0.13591$.

c) $P(X \geq x) = P\left(Z > \frac{x-6000}{100}\right) = 0.95$.

$\frac{x-6000}{100} = -1.65 \Rightarrow x = 5835$.

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4-53/449. The line width for semiconductor manfr is assumed to be normally distributed with mean of 0.5 μm & a std dev. of 0.05 μm.

a) What is the prob. that a line width is > 0.62 μm?

b. What is the prob that " " " b/w 0.47 & 0.63 μm?

c) The line width of 90% of samples is below what value?

a. $P(X>0.62) = P\left(Z > \frac{0.62-0.5}{0.05}\right) \Rightarrow P(Z>2.4)$

$= 1 - P(Z<2.4) = 0.0082.$

b. $P(0.47<X<0.63) = P\left(\frac{0.47-0.5}{0.05} < Z < \frac{0.63-0.5}{0.05}\right)$

$= P(-0.6 < Z < 2.6) = P(Z<2.6) - P(Z<-0.6)$

$= 0.99534 - 0.27425 \Rightarrow 0.72109.$

c) $P(X<x) = P\left(Z < \frac{x-0.5}{0.05}\right) = 0.90.$

$\frac{x-0.5}{0.05} = 1.28 \Rightarrow x = 0.564$

4-56.

a) $P(X>0.5) = P\left(Z > \frac{0.5-0.4}{0.05}\right) = P(Z>2) = 0.02275.$

b) $P(0.4<X<0.5) = P\left(\frac{0.4-0.4}{0.05} < Z < \frac{0.5-0.4}{0.05}\right)$

$= P(0<Z<2) = 0.47725$

c. $P(X>x) = 0.9 \rightarrow P\left(Z > \frac{x-0.4}{0.05}\right) = 0.90.$

$\therefore \frac{x-0.4}{0.05} = -1.28 \Rightarrow x = 0.336.$

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4.57/4.54 The speed of a file transfer from a server on campus to a personal computer at a student's home on a weekday evening is normally distributed with a mean of 60 kilobits/sec & std. devn. of 4 kBits/sec.

a) What is the prob. that the file will transfer at a speed of 70 kBits/sec or more?

b. What is the prob that file will transfer at a speed of < 58 kb/sec?

c. If the file is 1 MB, what is the avg time it will take to transfer the file?

a) $P(X>70) = P\left(Z > \frac{70-60}{4}\right) = 1 - P(Z<2.5)$

$= 1 - 0.99379 = 0.00621.$

b. $P(X<58) = P\left(Z < \frac{58-60}{4}\right) = P(Z<-0.5)$

$= 0.308538.$

c. 1,000,000 bytes * 8 bits/byte = 8,000,000 bits.

$\frac{8,000,000}{60,000} = 133.33$ secs.

4.61/4.58 The life of a semiconductor lasers at a constant power is normally distributed with a mean of 7000 hrs and a std. devn of 600 hrs.

a) What is the prob. that a lasers fails before 5000 hrs?

b) What is the life in hrs that 95% of lasers exceed?

c) If 3 lasers are used in a pdt & they are assumed to fail independently, what is the prob that all 3 are still operating after 7000 hrs.

a. $P(X<5000) = P\left(Z < \frac{5000-7000}{600}\right)$

$= P(Z<-3.33) = 0.00043$

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b. $P(X>x)=0.95 \therefore P\left(Z>\frac{x-7000}{600}\right)=0.95$ & $\frac{x-7000}{600}=-1.64$

$\Rightarrow x=6016$

c. $P(X>7000)=P\left(Z>\frac{7000-7000}{600}\right)=0.5$.

$P(3 \text{ lasers operating after } 7000 \text{ hrs}) = \left(\frac{1}{2}\right)*\frac{1}{2}*\frac{1}{2}=\frac{1}{8}$

Normal Approximation to the Binomial and Poisson Distributions

If X is a binomial random variable with parameters n & p.

$$Z=\frac{X-np}{\sqrt{np(1-p)}} \quad \text{—— (1)}$$

is a std normal random variable. To approximate a binomial prob. with a normal distn. a continuity correction is applied as follows.

$$P(X\leq x)=P(X\leq x+0.5)\cong P\left(Z\leq \frac{x+0.5-np}{\sqrt{np(1-p)}}\right)$$

&

$$P(x\leq X)=P(x-0.5\leq X)\cong P\left(\frac{x-0.5-np}{\sqrt{np(1-p)}}\leq Z\right)$$

The approx is good for $np>5$ & $n(1-p)>5$.

Recall that for a binomial variable X, $E(X)=np$ & $V(X)=np(1-p)$. Consequently, the eqn (1) is nothing more than the formula for standardizing random variable X.

The normal approx. is good when n is large relative to p.

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A way to remember the approx is to write the prob in terms of $\leq$ or $\geq$ then add or subtract 0.5 correction factor to make prob. greater.

Binomial distn is a satisfactory approx to hypergeometric distn when n, the sample size is small relative to N, size of popln.

A rule of thumb is that the binomial approx is effective if $n/N < 0.1$.

For a hypergeometric distn. p is defined as $p = K/N$. p is interpreted as the no of successes in the popln. ∴ Normal distn can provide effective approx of hypergeometric prob. when $n/N < 0.1$, $np > 5$ & $n(1-p) > 5$.

hypergeometric distn $\xrightarrow{\frac{n}{N} < 0.1}$ binomial distn $\xrightarrow{\substack{np>5 \\ n(1-p)>5}}$ Normal distn

Poisson distn. was developed as the limit of a binomial distn. as no. of trials increased to infinity. Hence it should not be suprising to find normal distn can also be used to approx. prob. of a poisson R.V. If X is a poisson r.v with $E(X) = \lambda$ & $V(X) = \lambda$.

$$Z = \frac{X - \lambda}{\sqrt{\lambda}}$$ is $\approx$ a std normal R.V.

The approx. is good for $\lambda > 5$.

Problems

1. Suppose that X is a binomial R.V with $n = 200$ & $p = 0.4$
   a) Approx the prob that X is ≤ 70.
   b) " " " " $X > 70$ & < 90.
   c) " " " $X = 80$.

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$E(X) = 200*(0.4) = 80$. $V(X) = 200*0.4*0.6 = 48$

$\sigma(X) = 6.93$

a. $P(X \leq 70) \cong P\left(Z \leq \frac{70-80}{\sqrt{48}}\right) = P(Z \leq -1.44) = 0.0749$

b. $P(70 < X \leq 90) \cong P\left(\frac{70-80}{\sqrt{48}} < Z \leq \frac{90-80}{\sqrt{48}}\right) =$

$P(-1.44$

a) $P(X \leq 70.5)$ $z = \frac{x-\mu}{\sigma} = \frac{70.5-80}{6.93} = -1.37$.

$P(Z \leq -1.37) = 0.08534$

b. $P(70 < X \leq 90) = P(70 \leq X \leq 89.5)$

for 70.5 $z = -1.37$ for 89.5 $z = (x-\mu)/\sigma$

$= (89.5-80)/6.93 = 1.37$.

$P(-1.37 \leq Z \leq 1.37) = P(Z \leq 1.37) - P(Z \leq -1.37)$

$= 0.91466 - 0.08534 = 0.82932$.

c. $P(X = 80) = P(79.5 \leq X \leq 80.5)$

$= P(-0.072 < Z < 0.072) = 0.0558$.

2. Assume that the number of asbestos particles in a squared meter of dust on a surface follows a Poisson dist with a mean of 1000. If a squared meter of dust is analysed, what is the prob that less than 950 particles are found?

$$P(X \leq 950) = \sum_{x=0}^{950} \frac{e^{-1000}\,1000^{x}}{x!} \rightarrow \text{computational difficulty.}$$

$$P(X \leq x) = P\left(Z \leq \frac{950-1000}{\sqrt{1000}}\right) = P(Z \leq -1.58) = 0.057$$

What is the prob that 10 cm^2 of dust contains more than 10,000 particles

$\lambda = 10*(1000) = 10,000$ $E(X) = \lambda = 10^4$ $V(X) = \lambda = 10^4$

$P(X > 10,000) \cong P\left(Z > \frac{10,000-10000}{\sqrt{10000}}\right) = P(Z > 0) = 0.5$

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3. 15% of parts fail on being subjected to a qualification test. The parts fail independently of other parts. 300 parts are tested every day. Compute the foll.

a) Prob that b/w 35 & 45 parts fail in a day (excluding 35 & 45)

b) Prob that there are atmost 25 part failures in a day

c) Of 10 consecutive days when parts are tested, what is the prob that there were exactly 3 days with 25 or less part failures

$p = 0.15 \quad n = 300. \quad 300 * 0.15 = 45 \quad V(x) = 300 * 0.15 * 0.85 = 38.25 = 6.185$

a) $P(35.5 \le Z \le 44.5) = P\left(\frac{35.5-45}{6.185} \le Z \le \frac{44.5-45}{6.185}\right)$

$= P(-1.536 \le Z \le -0.08) = 0.468 - 0.061 = 0.407$

b) $P(Z \le 25) = P(Z \le 24.5) = P\left(\frac{24.5-45}{6.185}\right) = -3.31 = 0.000434$

c) D days with ≤ 25 failures $\sim$ Bin (10, .0004)

$P(D=3) = \binom{10}{3} (0.0004)^3 (0.9996)^7 =$

4. A high volume printer produces minor print quality errors on a test pattern of 1000 pages of text acc. to poisson distn. with a mean of 0.4/pg.

a) Why are the no. of errors on each pg independent random variables?

b) What is the mean no. of pages with errors (1 or more)?

c) Approx. the prob. that more than 350 pgs contain errors (1 or more)

Let X rep. the no. of minor errors on a test pattern of 1000 pgs of text. X is Poisson R.V with mean 0.4/pg.

a) The no. of errors/pg are Random variables. The assumption that the occurrence of an event in a subinterval in a Poisson process is independent of events in other subintervals.

$\Rightarrow$ that no of events in disjoint intervals are independent. The pages are disjoint intervals $\Rightarrow$ consequently error count/pg are independent

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b) $P(X=0) = \frac{e^{-0.4}\,0.4^0}{0!} = 0.670$

$P(X \geq 1) = 1 - P(X=0) = 1 - 0.670 = 0.330$

The mean no. of pgs with 1 or more errors is

$1000 * (0.330) = 330$ pgs

c) Let Y be no of pages with errors.

$P(Y > 350) \approx P\left(Z \geq \frac{350 - 330}{\sqrt{1000 * (0.33)(0.67)}}\right)$

$P(Z \geq 1.35) = 1 - P(Z \leq 1.35)$

$= 1 - 0.9115 = 0.0885$

5. The % of people exposed to a bacteria who become ill is 20%. Assume that people are independent. Assume that 1000 people are exposed to the bacteria. Approximate each of all

a) The prob that more than 225 become ill.

b) The prob that b/w 175 & 225 become ill.

c) The value such that prob. that the no. of people that become ill exceeds the value is 0.01.

$E(X) = 1000\,(0.2) = 200$ and $V(X) = 1000 * 0.2 * 0.8 = 160$.

a) $P(X \geq 225.5) = 1 - P\left(Z \leq \frac{225.5 - 200}{\sqrt{160}}\right) = 1 - P(Z \leq 2.015)$

$= 1 - 0.9777 =$

$= 0.0223$

b. $P(175.5 \leq X \leq 224.5) \approx P\left(\frac{175.5 - 200}{\sqrt{160}} \leq Z \leq \frac{224.5 - 200}{\sqrt{160}}\right)$

$= P(-1.93 \leq Z \leq 1.93)$

$= P(Z \leq 1.93) - (Z \leq -1.93)$

$= 0.973197 - 0.026803 = 0.946394$

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c) If $P(X>x) = 0.01$ then

$$\left[Z > \frac{(x+0.5)-200}{\sqrt{160}}\right] = 0.01.$$

$$\therefore \frac{(x+0.5)-200}{\sqrt{160}} = 2.33$$

$$\therefore x+0.5 = 200 + 29.47$$

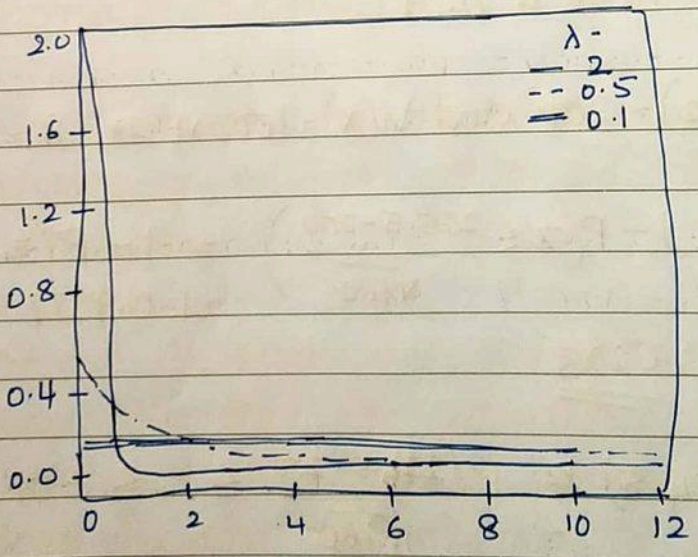
$$x = 229.5 + 0.5 = 230$$

EXPONENTIAL DISTRIBUTION.

Defn. The random variable X that equals the distance b/w successive counts of a Poisson process with mean $\lambda > 0$ is an exponential random variable with parameter λ. The pdf of X is

$$f(x) = \lambda e^{-\lambda x} \text{ for } 0 \le x < \infty.$$

The exponential distn. obtains its name from the exponential fn. in the pdf. Plots of exponential distn for selected values of λ are shown below



If the random variable X has an exponential distn with parameter λ,

$$\mu = E(X) = \frac{1}{\lambda} \text{ and } \sigma^2 = V(X) = \frac{1}{\lambda^2}$$

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$F(x) = 1 - e^{-\lambda x}, \quad x \geq 0.$ DATE

It is imp. to use consistent units in calculation of mean, variances involving exponential r.v.

Problems.

1. Suppose X has an exponential distn. with $\lambda = 2$. Det. the foll.
   a) $P(X \leq 0)$ b) $P(X \geq 2)$ c) $P(X \leq 1)$ d) $P(1 < X < 2)$
   e) Find the value of x such that $P(X < x) = 0.05$

a. $P(X \leq 0) = \int_0^0 \lambda e^{-\lambda x} dx = 0.$

b. $P(X \geq 2) = \int_2^\infty 2e^{-2x} dx = -e^{-2x}\Big|_2^\infty = e^{-4} = 0.0183.$

c. $P(X \leq 1) = \int_0^1 2e^{-2x} dx = -e^{-2x}\Big|_0^1 = 1 - e^{-2} = 0.8647.$

d. $P(1 < X < 2) = \int_1^2 2e^{-2x} dx = -e^{-2x}\Big|_1^2 = e^{-2} - e^{-4} = 0.1170.$

e. $P(X \leq x) = \int_0^x 2e^{-2t} dt = -e^{-2t}\Big|_0^x = 1 - e^{-2x} = 0.05 \Rightarrow x = 0.0256$

2. The time b/w calls to a plumbing supply business is exponentially distributed with a mean time b/w calls of 15 mins.
   i What is the probability that there are no calls within a 30 min interval?
   ii What is the prob that atleast one call arrives within a 10 min interval?
   iii. What is the " " the first call arrives within 5 and 10 mins after opening?
   iv. Det. the length of an interval of time such that the prob. of atleast one call in the interval is 0.90.

T ~ Exponential $(1/15)$

(i) $P(\text{No. calls in 30 min period}) = P(T \geq 30) = \int_{30}^\infty \frac{1}{15} e^{-\frac{x}{15}} dx = -e^{-\frac{x}{15}}\Big|_{30}^\infty$
$= e^{-\frac{1}{15} * 30} = e^{-2} = 0.1353.$

or X = calls per 30 mins period $\sim Poi(30 * \frac{1}{15})$

$P(X = 0) = \frac{e^{-2} 2^0}{0!} = 0.1353.$

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ii) P(at least 1 call in 10 min interval)

Define Y = no of calls in 10 min period $\sim Poi(10 * \frac{1}{15})$

$$P(Y \geq 1) = 1 - P(Y=0) = 1 - \frac{e^{-2/3}(2/3)^0}{0!} = 1 - e^{-2/3}$$

$$= 1 - 0.5134 = 0.4866$$

$= 1 - P(X>10) = 1 - P(x>10) \Rightarrow P(x>10) = -e^{-\frac{x}{15}}\Big|_{10}^{\infty} = e^{-2/3} = 0.5134$

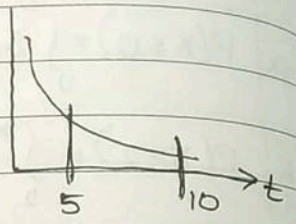
$\Rightarrow 1 - 0.5134 = 0.4866$

iii $P(5 \leq T \leq 10) = P(T \geq 5) - P(T \geq 10)$

$= e^{-1/15 * 5} - e^{-\frac{1}{15} * 10}$

$= 0.2031$

$= -e^{x/15}\Big|_5^{10} = e^{-1/3} - e^{-2/3} = 0.2031$



iv. Let the interval be t

Then w, no of calls in period t $\sim Poit(\frac{1}{15}t)$

$$P(W \geq 1) = 0.9 \Rightarrow 1 - P(W=0) = 0.9 \Rightarrow P(W=0) = 0.1$$

$$= \frac{e^{-\frac{1}{15}t}(\frac{1}{15}t)^0}{0!} \Rightarrow e^{-\frac{1}{15}t} = 0.1$$

$$t = -15 \ln 0.1 = 34.5 \text{ mins}$$

$P(X<x) = 0.9$ & $P(X<x) = -e^{-x/15}\Big|_0^x = 1 - e^{-x/15} = 0.9 \Rightarrow x = [\ln(0.1)] * 15$

$= 34.54$ mins.

3. Inter arrival times in minutes b/w successive customers in a hotel is exponentially distributed with a mean of 2mins.

a) Compute the prob. that IAT b/w 2 customers is atmost 2mins.

b) What is the prob. that at least 2 customers arrive in one min?

c) What is the mean time till every 3rd customer arrives?

d) What is the prob. of no customers arriving in a span of 3mins?

$\lambda = 1/2 = 0.5$

T $\sim$ Exponential (0.5)

a) $P(T \leq 2 \text{ min}) = 1 - e^{-\lambda x} = 1 - e^{-2 * .5} = 0.6321.$

b) X = no. of customers/min $\sim$ Poiss(0.5)

$P(X \geq 2) = 1 - [P(X=0) + P(X=1)$

$= 1 - e^{-.5} - e^{-.5} * .5 = 1 - 0.6065 - 0.3033$

$= 1 - 0.9098 = 0.0901$

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c) Time to 3rd customer, $T' \sim Erlang(3, .5)$

$$E(T') = \frac{3}{.5} = 6 \text{ mins}.$$

d) Y, no of customers every 3 mins $\sim Poi(1.5)$ $\quad 3*0.5$

$$P(Y=0) = \frac{e^{-1.5}(1.5)^0}{0!} = 0.2231$$

4. Suppose the counts recorded by a Geiger counter follow a poisson pr. with an average of 2 counts/min.

a) What is the prob that there are no counts in a 30sec interval.

b) " " " " " the first count occurs in < 10secs.

c) " " " " " " " " " b/w 1 & 2 mins after start up?

Let X denote the time until 1st count. Then X is an exponential random variable with $\lambda = 2$ counts/min.

a) $P(X > 0.5) = \int_{0.5}^{\infty} 2 * e^{-2x}\, dx = -e^{-2x}\Big|_{0.5}^{\infty} = e^{-1} = 0.3679$.

b) $P(X < \frac{10}{60}) = \int_0^{1/6} 2e^{-2x}\, dx = -e^{-2x}\Big|_0^{1/6} = 1 - e^{-1/3} = 0.2835$.

c) $P(1 < X < 2) = -e^{-2x}\Big|_1^2 = e^{-2} - e^{-4} = 0.1170$.

5. The time b/w arrivals of taxis at a busy intersection is exponentially distributed with a mean of 10 mins.

a) What is the prob that you wait longer than 1 hr for a taxi?

b) Suppose you have already been waiting for 1 hr for a taxi, what is the prob that one arrives within the next 10 mins.

Let X denote the time until the arrival of a taxi. Then X is an exponential r.v with $\lambda = 1/E(x) = 0.1$ arrivals/min.

a $P(X > 60) = \int_{60}^{\infty} 0.1 * e^{-0.1x}\, dx = -e^{-0.1x}\Big|_{60}^{\infty} = e^{-6} = 0.0025$.

b. $P(X < 10) = \int_0^{10} 0.1 e^{-0.1x}\, dx = -e^{-0.1x}\Big|_0^{10} = -e^{-1} + 1 = 0.6321$

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6. The dist. b/w major cracks in a highway follows an exponential distn with mean of 5 miles.

a) What is the prob. that there are no major cracks in 10 miles

b) " " " " " " 2 " " " " "

c) What is the std devn of the dist. b/w major cracks?

Let X denote the ~~no. of cracks in 10 miles of highway~~ distance b/w major cracts.

Then X is exponential r.v with

$$\lambda = \frac{1}{E(X)} = \frac{1}{5} = 0.2 \text{ cracks/mile}$$

a. $$P(X>10) = \int_{10}^{\infty} 0.2 e^{-0.2x} dx = -e^{-0.2x} \Big|_{10}^{\infty}$$

$$= e^{-2} = 0.1353$$

b) Let Y denote the no. of cracks in 10 miles of highway.

∵ the distance b/w cracks is exponential, Y is a Poisson R.V with $\lambda = 10 * 0.2 = 2$ cracks/10 miles.

$$P(Y=2) = \frac{e^{-2} * 2^2}{2!} = 0.2707.$$

c. $\sigma_X = 1/\lambda = 5$ miles

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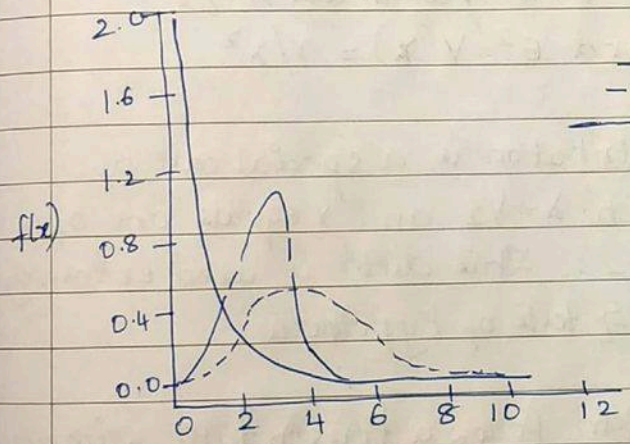
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ERLANG AND GAMMA DISTRIBUTIONS.

An exponential R.V describes the length until the first count is obtained in a Poisson process. A generalisation of the exponential distn is the length until r counts occur in a poisson process. The R.V that equals the interval length until r counts occur in a poisson process has an Erlang random variable.

The random variable X that equals the interval length until r counts occur in a poisson process with mean $\lambda > 0$ has an Erlang R.V with parameters λ and r. The pdf of X is

$$f(x) = \frac{\lambda^r x^{r-1} e^{-\lambda x}}{(r-1)!} \quad \text{for } x > 0 \text{ \& } r = 1, 2, \ldots$$



If X is an Erlang r.v. with parameters λ and r,

$$\mu = E(X) = r/\lambda \quad \text{and} \quad \sigma^2 = V(X) = r/\lambda^2$$

Gamma distribution – The Erlang distribution is a special case of gamma distn. If the parameter r of an Erlang R.V is not an integer but $r > 0$, the r.V has a gamma distn.

The gamma function is

$$\Gamma(r) = \int_0^\infty x^{r-1} e^{-x} dx, \quad \text{for } r > 0$$

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$\Gamma(r) = (r-1)\Gamma(r-1)$

$\Gamma(r) = (r-1)!$ If r is an +ve integer (as in Erlang)

$\Gamma(r) = (r-1)!$ $\quad \Gamma(1) = 0! = 1$

$\Gamma(1/2) = \pi^{1/2}$

The random variable X with pdf

$$f(x) = \frac{\lambda^r x^{r-1} e^{-\lambda x}}{\Gamma(r)}, \quad \text{for } x > 0$$

has a gamma r.v with parameter $\lambda > 0$ & $r > 0$.
If r is an integer, X has a Erlang distribution

If X is a gamma r.v with parameters λ & r
$\mu = E(X) = r/\lambda$ and $\sigma^2 = V(X) = r/\lambda^2$

The chi-squared distribution is a special case of gamma dist^n. in which $\lambda = 1/2$ and r equals one of the values $1/2, 1, 3/2, 2, \ldots$ This dist^n is used extensively in interval estimation & tests of hypothesis

1. Calls to a telephone system follow a Poisson dist^n. with mean of 5 calls/min.

a) What is the name applied to dist^n & parameter values of the time until the tenth call?

b) What is the mean time until the tenth call?

c) What is the mean time b/w the 9th & 10th calls?

a) The time until the tenth call is an Erlang R.V with $\lambda = 5$ calls/min & $r = 10$.

b) $E(X) = r/\lambda = 10/5 = 2$ mins.
$V(X) = r/\lambda^2 = 10/25 = 0.4$ mins.

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a poisson process is memoryless, the mean time is $r/\lambda = 1/5 = 0.2$ mins or 12 seconds

d. What is the prob that exactly 4 calls occur within 1 min?

e. If 10 separate 1 min intervals are chosen, what is the prob that all intervals contain more than 2 calls?

Let Y denote the no. of calls in 1 min. Then Y is a Poisson R.V with $\lambda = 5$ calls/min.

d) $P(Y=4) = \frac{e^{-5}5^4}{4!} = 0.1755$

e) $P(Y>2) = 1 - P(Y\leq 2) = 1 - \frac{e^{-5}5^0}{0!} - \frac{e^{-5}5^1}{1!} - \frac{e^{-5}5^2}{2!} = 0.8754$

Let W denote the no. of 1 min intervals out of 10 that contain more than 2 calls.

Because calls are a Poisson process, W is a binomial r.v with $n=10$ & $p=0.8754$

$\therefore P(W=10) = \binom{10}{10}(0.8754)^{10}(1-0.8754)^0 = 0.2643$

2. The time b/w failures of a lases in a cytogenics m/c is exponentially distributed with a mean of 2500 hrs.

a) What is the expected time until the second failure?

b) What is the prob. that the time until the 3rd failure exceeds 50,000 hrs?

exponential $E(X) = 2500 \Rightarrow \frac{1}{\lambda} = 2500 \Rightarrow \lambda = 0.0004$

a) Expected time until the 2nd failure $E(X) = r/\lambda = 2/0.0004$
$= 50,000$ hrs.

b. N = no of failures in 50,000 hrs.

$E(N) = \frac{50000}{25000} = 2$

$P(N\leq 2) = \sum_{k=0}^{2} \frac{e^{-2}(2)^k}{k!} = 0.6767.$

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3. The time b/w process problems in a manfc. line is exponentially distributed with mean of 30 days.

a) What is the expected time until the 4th problem?

b) " " " prob that the time " " " " exceeds 120 days?

Let X denote the time in days until the 4th problem. Then X has an Erlang distribution with $r = 4$ & $\lambda = 1/30$ problems / day.

(a) $E(X) = \frac{4}{1/30} = 120$ days.

b) Let Y denote the no of problems in 120 days. Then Y has a poisson R.V with $\lambda = 4$ problems / 120 days.

$$P(Y \geq 4) = \frac{e^{-4}4^0}{0!} + \frac{e^{-4}4^1}{1!} + \frac{e^{-4}4^2}{2!} + \frac{e^{-4}4^3}{3!} = 0.4335$$

4. Use the gamma fn. to evaluate the foll

a) $\Gamma(6) = 5! = 120$.

b) $\Gamma\left(\frac{5}{2}\right) = \frac{3}{2}\Gamma\left(\frac{3}{2}\right) = \frac{3}{2} * \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{3}{4} * \pi^{1/2} = 1.32934$.

c) $\Gamma\left(\frac{9}{2}\right) = \frac{7}{2}\Gamma(7/2) = \frac{7}{2} * \frac{5}{2} * \frac{3}{2} * \frac{1}{2}\Gamma(1/2)$

$= \frac{105}{16} * \pi^{1/2} = 11.6317$

5. The TBA of customers at an automatic teller m/c is an exponential r.v with a mean of 5mins?

a) What is the prob. that more than 3 customers arrive in 10 mins?

b) What is the prob that the time until 5th customer arrives

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is less than 15 mins?

a. Let X denote the no of customers that arrive in 10 mins. Then X is a poisson r.v with $\lambda = 0.2$ arrivals/min = 2 arrivals per 10 min.

$$P(X>3) = 1 - P(X \le 3) = 1 - \left[\frac{e^{-2}2^0}{0!} + \frac{e^{-2}2^1}{1!} + \frac{e^{-2}2^2}{2!} + \frac{e^{-2}2^3}{3!}\right]$$

$$= 0.1429$$

b. Let Y denote the no. of customers who arrive in 15 mins. $\lambda = 0.2$ arrivals/min $\therefore$ for 15 mins $\lambda = 3$.

$$P(Y>3) = 1 - P(Y \le 3) = 1 - \left[\frac{e^{-3}3^0}{0!} + \frac{e^{-3}3^1}{1!} + \frac{e^{-3}3^2}{2!} + \frac{e^{-3}3^3}{3!}\right]$$

$$= 1 - e^{-3}\left[1 + 3 + 9/2 + 27/6\right] = 0.1847$$

Weibull distribution

It is used to model the time until failure of many physical systems.

The random variable X with pdf

$$f(x) = \frac{\beta}{\delta}\left(\frac{x}{\delta}\right)^{\beta-1} \exp\left[\left(-\frac{x}{\delta}\right)^{\beta}\right] \text{ for } x>0.$$

is a weibull r.v with scale parameter $\delta > 0$ & shape parameter $\beta > 0$.

If $\beta = 1$, the weibull distribution is identical to exponential distribution.

The cdf fn of X which has weibull dist with parameters δ & β are

$$F(x) = 1 - e^{-\left(\frac{x}{\delta}\right)^{\beta}} \qquad 1 - e^{-\left(\frac{x}{\delta}\right)^{\beta}}$$

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$$\mu = E(X) = \delta \Gamma\left(1 + \frac{1}{\beta}\right)$$

$$\sigma^2 = V(X) = \delta^2 \Gamma\left[1 + \frac{2}{\beta}\right) - \delta^2 \left[\Gamma\left(1 + \frac{1}{\beta}\right)\right]^2$$

1. The time to failure (in hrs) of a bearing in a mechanical shaft is modeled as a weibull r.v with $\beta = 0.5$ & $\delta = 5000$ hrs. Det the mean time until failure & det the prob that a bearing lasts at least 6000 hrs

$$E(X) = 5000\,\Gamma\left(1 + \tfrac{1}{0.5}\right) = 5000\,\Gamma(3) = 5000 \times 2!$$

$$= 10{,}000 \text{ hrs.}$$

$$P(X > 6000) = 1 - F(6000) = \exp - \left[\left(\frac{6000}{5000}\right)^{0.5}\right]$$

$$= e^{-1.095} = 0.334$$

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